



MGM's College of Engineering, Nanded

Department of Computer Science & Engineering Academic Year 2025-26

"Edu2 Job – Predicting Job Roles from Educational Background"

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Introduction:

This project, "Edu2Job – Predicting Job Roles from Educational Background", is an intelligent career guidance platform that predicts suitable job roles for students based on their profile, skills, projects, and experience. It uses Machine Learning to analyze student data and recommend top career roles with confidence scores. The system also analyzes resumes, identifies skill gaps, and suggests learning resources to help students improve their employability and achieve their career goals. By combining data-driven predictions with AI-powered insights, Edu2Job aims to bridge the gap between education and industry requirements and support students in making informed and confident career decisions.

System Architecture:

The Edu2Job system consists of a Streamlit-based user interface, backend services using FastAPI, ML models, and a MongoDB database.

The system handles user authentication, profile management, resume analysis, career role prediction, skill gap analysis, course recommendations, and performance tracking.

It analyzes user data and provides personalized career guidance through a modular architecture ensuring scalability, security, and smooth user experience.

SYSTEM ARCHITECTURE



Methodology:

The Edu2Job system follows a structured methodology that starts with requirement gathering and data collection.

Student data, including academic background, skills, projects, and experience, is preprocessed and stored in the database. Machine learning models are trained and evaluated to predict suitable job roles.

The system performs resume analysis using NLP, identifies skill gaps, and recommends courses and learning resources. User performance is tracked and visualized through dashboards. Continuous feedback helps improve prediction accuracy and recommendations, ensuring a personalized and effective career guidance experience for every user.

METHODOLOGY



Conclusion:

The Edu2Job project successfully builds an AI-powered career guidance platform that predicts job roles from educational background. It analyzes resumes, identifies skill gaps, and recommends learning resources to enhance employability. The system is designed to be scalable, efficient, and user-friendly for students and institutions.

Maps to PO's

PO-1	PO-6	PO-11	PO-16
PO-2	PO-7	PO-12	PO-17
PO-3	PO-8	PO-13	PO-18
PO-4	PO-9	PO-14	PO-19
PO-5	PO-10	PO-15	PO-20

Name of Students

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